Detecting Autism Spectrum Disorder via Facial Expressions and fMRI

ASD and it’s Implications

CONTENTS

01 Introduction

02 Objectives

03 Approaches

04 Architectures

05 Platform UI/UX

06 Future Work

Introduction

Introduction

- What is ASD?

A Neuro-developmental disorder

- It causes challenges social interactions, communication, and repetitive behavior

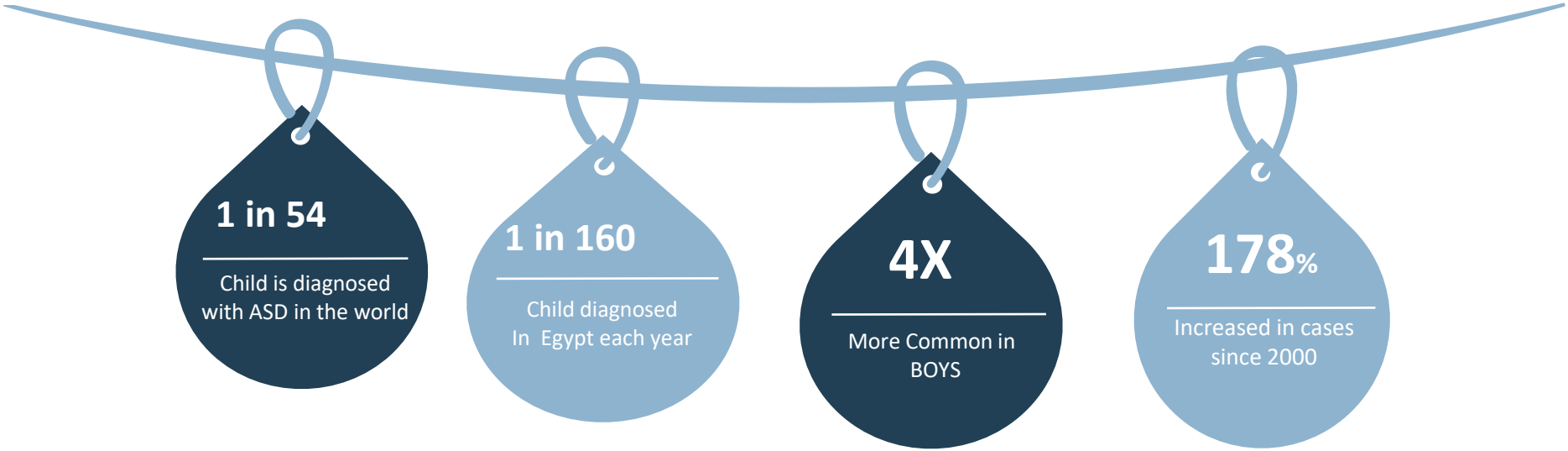
- Autism Spectrum

It's symptoms and severity vary widely among individuals

- Early diagnostics is key

It can be crucial for timely intervention and better development outcomes

Introduction

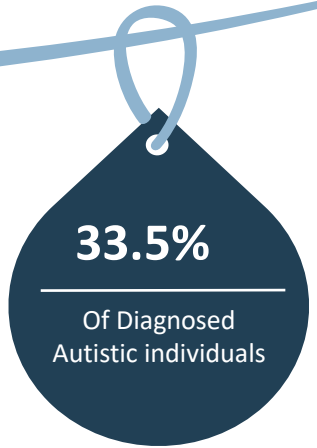


ASD Statistics

Reference: [How Common is Misdiagnosed Autism?]

Introduction

The problem is, MISDIAGNOSIS!
As 33.5% are being diagnosed
with another conditions.



ASD Statistics

Reference: https://link.springer.com/article/10.1007/S00406-020-01189-W

Objective

What is our project, and Why?

Objectives

Assist Doctors

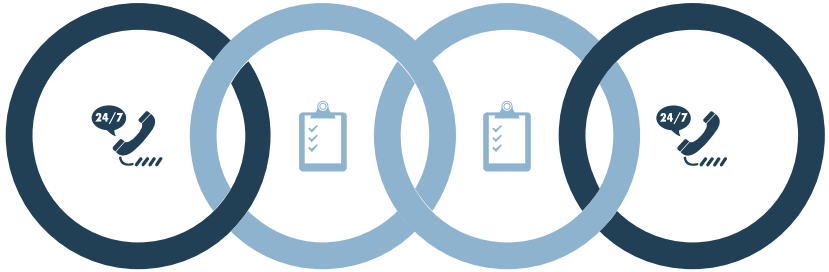
01

Providing an assistive tool to increase diagnostics accuracy, and reduce count of misdiagnoses.

Help Community

04

Raising awareness on what is autism, how does it affect people and understand autistic people more.



Assist Patients

03

Being diagnosed earlier, can lead to timely intervention which can improve the individual's skills.

Assist Families

02

Assist families with early diagnostics, tip & tricks on how to deal with autistic individuals, and provide further assistance.

Objectives

Educate and Raise Awareness

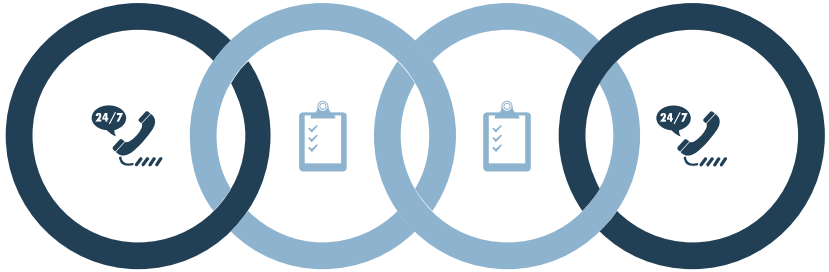
05

Train doctors on advanced diagnostic methods Support families with resources and guidance.

Collaborate for Impact

08

Partner with hospitals and autism organizations Integrate advanced tools through collaborations.



Improve Accessibility

06

Ensure tools are affordable and easy to use Expand availability to underserved areas..

Enhance Technology

07

Improve diagnostic accuracy using AI Continuously refine models with real-world data.

Approaches

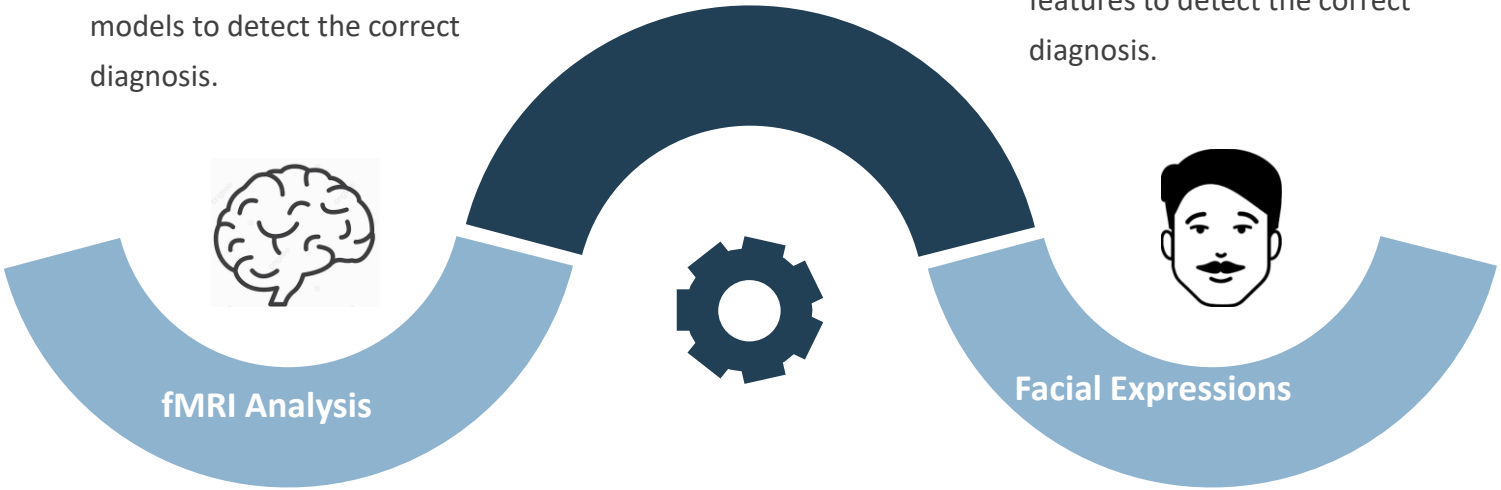
How will we reach our goal?

Approaches

Project Mechanism

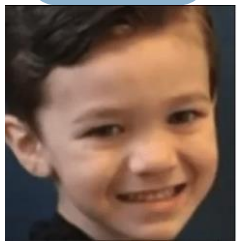
By using Brain Scans images - function MRI – in Neural Network models to detect the correct diagnosis.

By using facial expressions and features to detect the correct diagnosis.



Approaches

Datasets Used



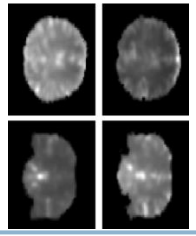
Preview of Facial Expressions Data

Source of Data:

Kaggle

Two different datasets used

Explained further in the upcoming parts.



Preview of fMRI Data

Source of Data:

ABIDE

Architectures

Data, Methods and Models.

Architectures

Data, Methods and Models.

Facial Expressions

Dataset Overview for Autism Classification (facial expression)

Source

Kaggle Image Dataset

Autism_Image_Data | Kaggle

Training Data

Total Training Images: 2540

Autistic Images: 1270 (50%)

Non-Autistic Images: 1270 (50%)

Testing Data

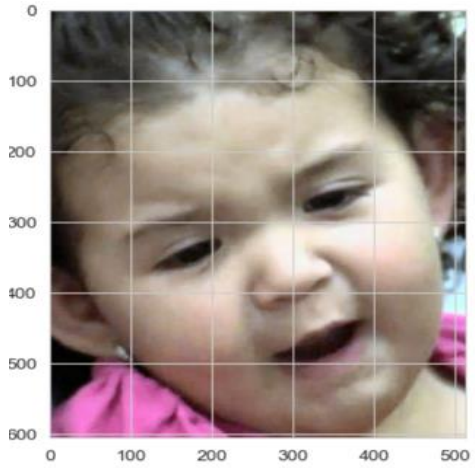
Total Test

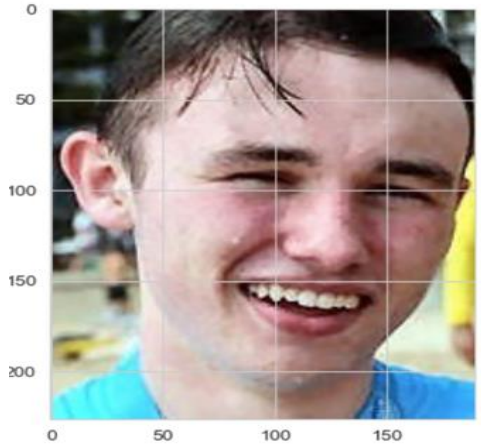
Images: 300

Image Dimensions

Varying dimensions

across images





Main Libraries Used

1 cv2 (OpenCV)

Read and resized images to 256x256 pixels for consistent input to the VGG16 model.

2 tensorflow

Built, compiled, and trained the VGG16 model for facial expression classification.

3 PIL (Pillow)

Handled basic image manipulation and transformations during preprocessing.

4 pandas

Managed image file paths, labels, and organized prediction results.

5 matplotlib

Visualized images, plotted accuracy/loss curves, and displayed confusion matrices.

6 sklearn (scikit-learn)

Evaluated model performance using accuracy scores, classification reports, and confusion matrices.

Architecture

Image Preprocessing and Data Augmentation

Standardization

Standardizes image size and format (e.g., 256x256 pixels).

Array Creation

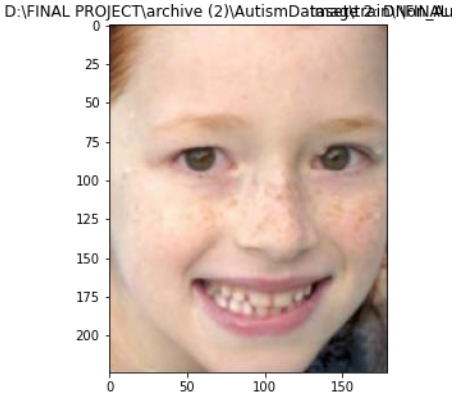
Converts images into numerical arrays suitable for model input.

Augmentation

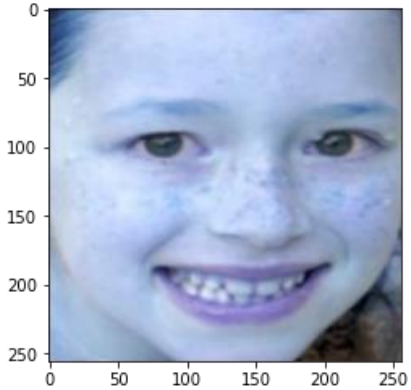
Applies random transformations (rotation, flipping, scaling, shifting).

Preprocessing and Augmentation Phase:

ORIGINAL IMAGE

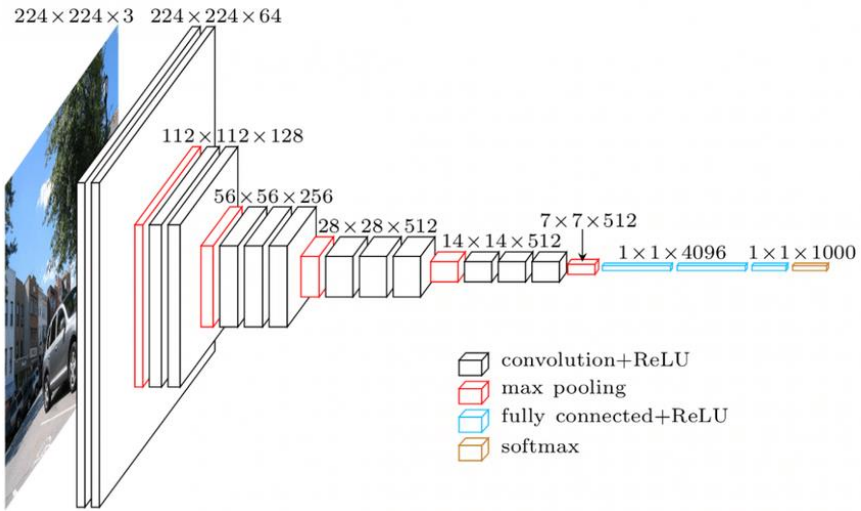


Preprocessing

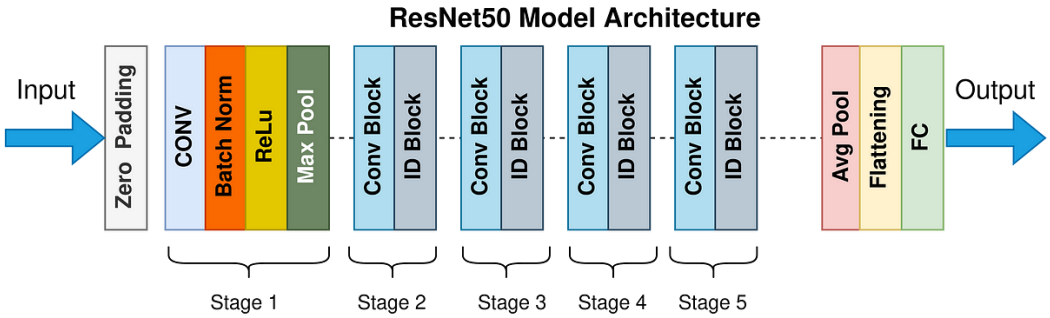


Augmentation

VGG-16 Model



ResNet-50 Model



Model Architecture

Title: deep convolutional neural network model introduced by visual geometry group at oxford university (VGG16).

input layer:

Input shape of image(256x256x3)

Convolutional layer:

13 convolutional layers distributed across 5 block, With ReLU Activation Layer.

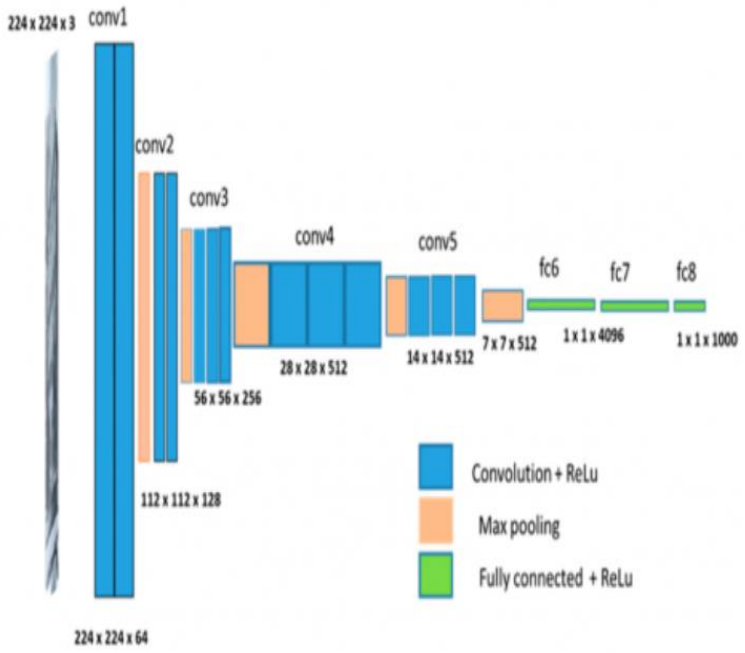
Pooling layer:

Max-pooling layers 2x2 (reduces dimensions)

Dropout: Prevent overfitting (50% in dense layers).

Fully Connected Layers:

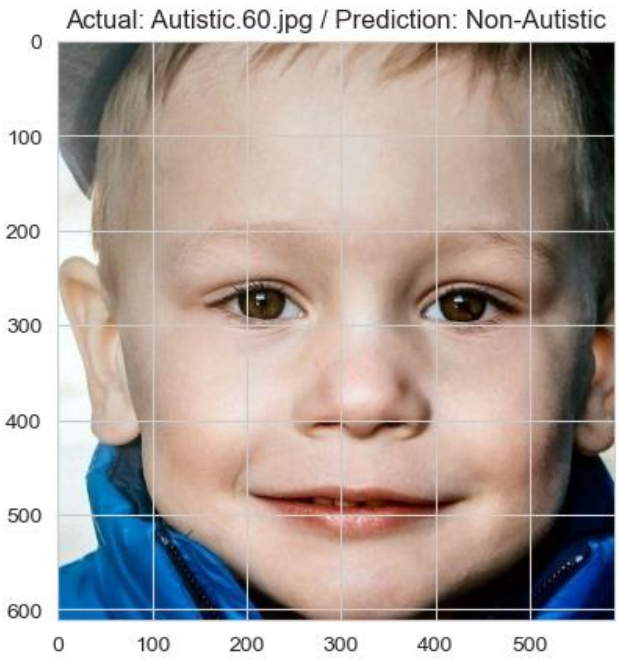
- Flatten →Dense(128) → Dense(1).
- Final Layer: Sigmoid activation for binary classification.



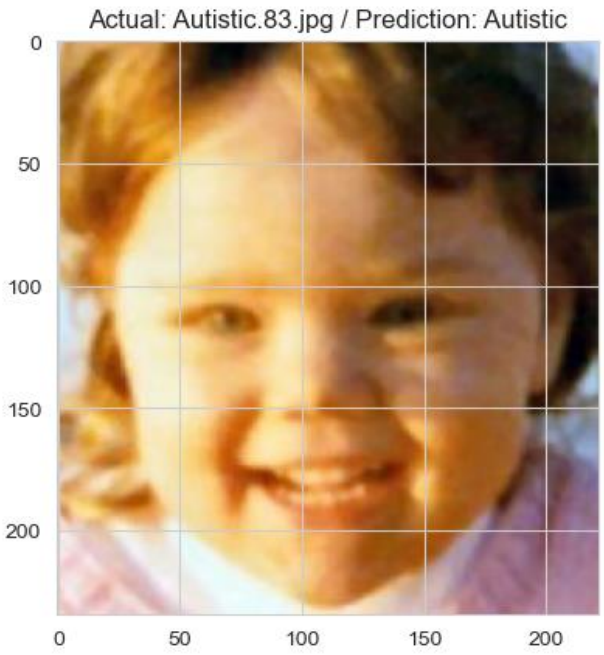
Training & Evaluation

- **Training**:
  - **Optimizer**: Adam.
  - **Loss**: Binary Cross-Entropy.
  - **Metrics**: Accuracy, Confusion Matrix, Classification Report.
  - **Callbacks**:
    - Early Stopping (Stop training if no improvement).
    - Model Checkpoint (Save the best model).

Training And Validation Accuracy



Example of False Negative



Example of True Positive

Comparison of ResNet50 and VGG16 for Autism Classification

- **Models Used**:
  - **ResNet50**
  - **VGG16**

- **Objective**:
  - Classify images to determine if a person has autism or not.

- **Results**:
  - **VGG16**: Achieved **75% validation accuracy**
  - **ResNet50**: Achieved **59% accuracy**

- **Conclusion**:
  - VGG16 outperforms ResNet50 in classifying autism images with higher accuracy.

Architectures

Data, Methods and Models.

fMRI

Architecture - Data

NIfTI

format is a widely used file format for storing neuroimaging data. It is specifically designed for volumetric data, such as functional MRI (fMRI)

bandpass filtering

Bandpass filtering is a preprocessing step applied to fMRI time series data to retain signals within a specific frequency range while removing signals outside this range

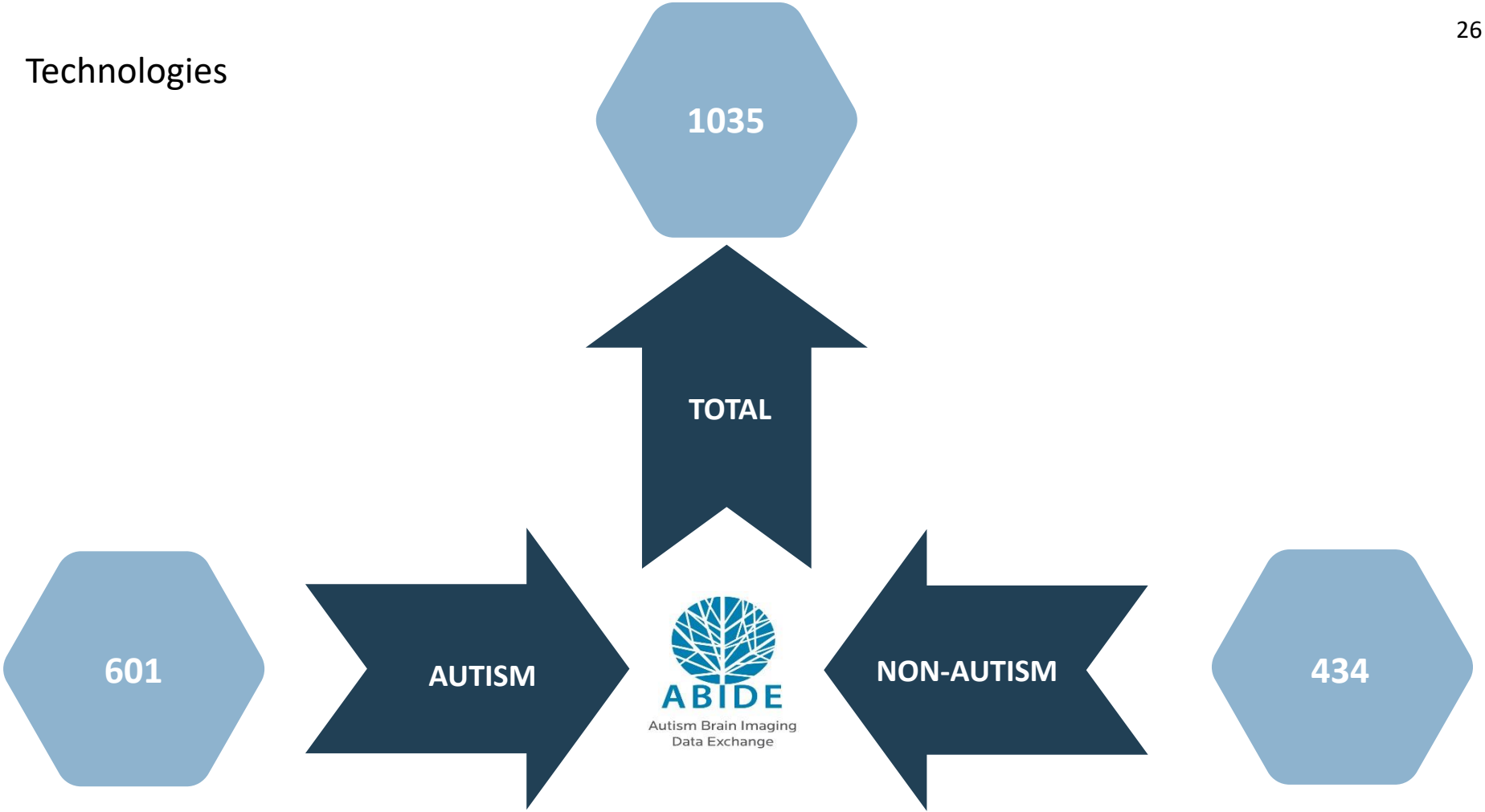
Amplitude of Low-Frequency Fluctuations (ALFF)

It is particularly useful for understanding the physiological underpinnings of brain function and detecting abnormalities in neurological or psychiatric conditions.

Configurable Pipeline for the Analysis of Connectomes (CPAC)

platform designed for analyzing neuroimaging data, particularly functional MRI (fMRI).

Technologies



Architecture - Data

ABIDE
Autism Brain Imaging Data Exchange

ASD Focused

It’s a collection and exchange of data focusing mainly on ASD.

Age Diverse

Contains diverse sample of different ages

Modularity

Contains both fMRI and sMRI, our main focus was on the fMRI images.

Phenotypic Date

Contains extensive phenotypic data for the explanation of the images

Sample Size

Total of 1035 Image,
601 Autistic individual, and 434 control ones.

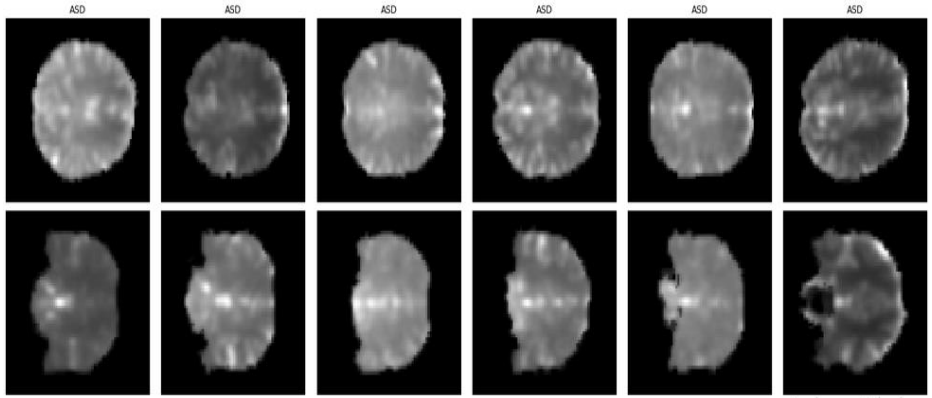
fMRI 3D-CNN Model

Goal: Classify brain scans into ASD (Autism Spectrum Disorder) or Control.

Input: 3D brain scan images (64×64×64×1).

Output: Binary classification (ASD vs. Control).

Tools Used: Python, TensorFlow, Keras, NumPy, and Nibabel.



(a) Autistic Image

(b) Non - Autistic Image

Model Architecture

Title: 3D Convolutional Neural Network (CNN)

•**Convolutional Layers**: Extract 3D spatial features using 3×3×3 filters.

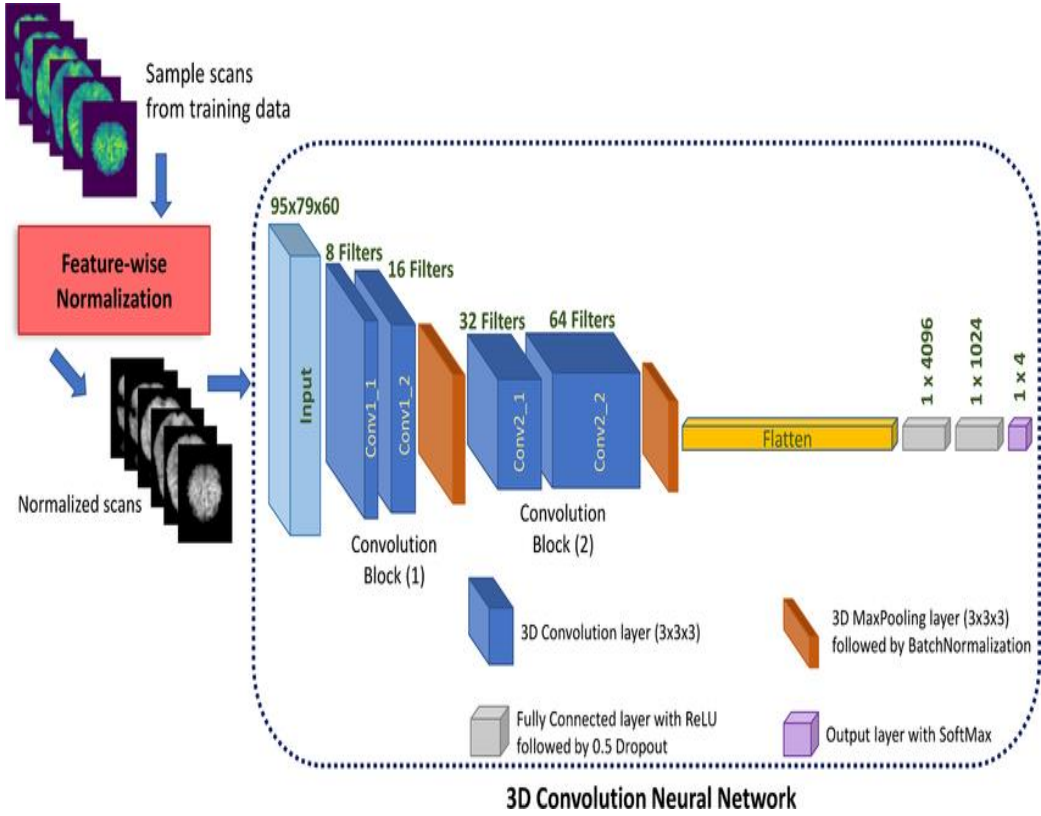
•**Pooling Layers**: Reduce dimensions and retain key features (2×2×2)

•**Batch Normalization**: Stabilize and speed up training.

•**Dropout**: Prevent overfitting (25% in convolutional layers, 50% in dense layers).

•**Fully Connected Layers**:

- Flatten → Dense(512) → Dense(256) → Dense(1).
- Final Layer: Sigmoid activation for binary classification.



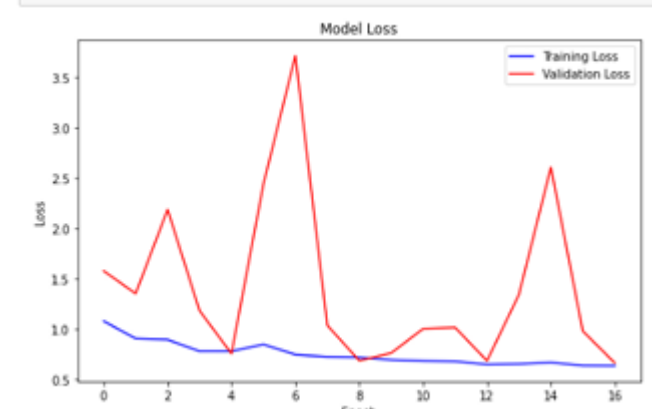
Training & Evaluation

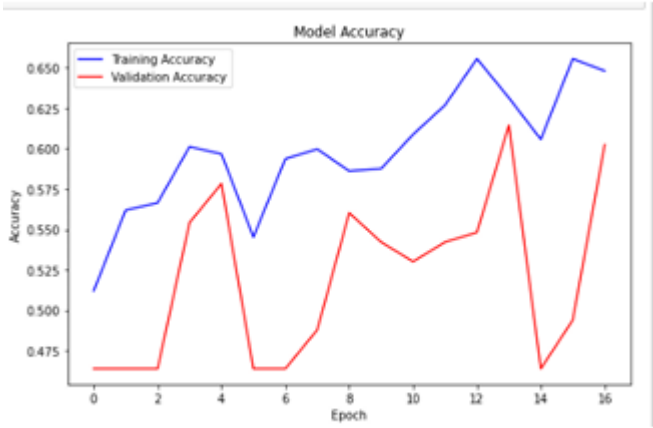
- **Training**:
  - **Optimizer**: Adam.
  - **Loss**: Binary Cross-Entropy.
  - **Metrics**: Accuracy and AUC (Area Under Curve).
  - **Callbacks**:
    - Early Stopping (Stop training if no improvement).
    - Model Checkpoint (Save the best model).
    - Live Training Visualization (Real-time accuracy and loss tracking).
- **Evaluation**:
  - **Metrics**: Confusion Matrix, ROC Curve, and Classification Report.

Model Results & Insights

- **Accuracy Achieved**: 68%
- **Evaluation Metrics**:
  - **ROC-AUC**: Demonstrates the model's ability to distinguish ASD vs. Control.
  - Confusion Matrix: Visual breakdown of predictions vs. actual.
- **Outputs**:
  - Binary Classification:
    - **1**: Autism
    - **0**: Control
- **Key Insight**: Moderate accuracy indicates potential for improvement with more data or hyperparameter tuning.

- Loss/accuracy graph



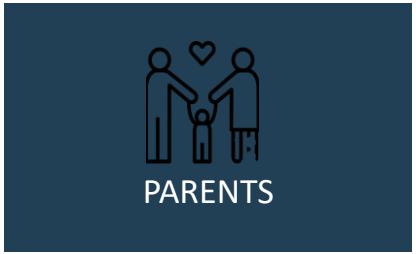


Platform UI/UX

Platform and Designs.

UI/UX

Targeted Audience



Reduce misdiagnosis

Results revealed a 33.5% misdiagnoses in autistic individuals.

Be Their Guide

lack of awareness and understanding of their children's unique learning needs.

UI/UX

ADD "**A**utism **D**isorder **D**etector"

GUI Platform Approach

Website

- Allows uploads of multi-formats
  - Allows history of uploads

Mobile App

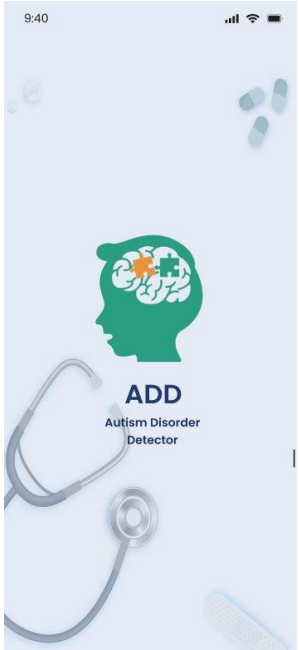
- Camera Integration
- Push Notifications

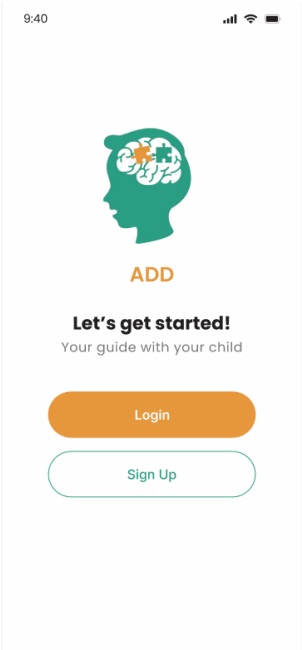
Desktop GUI

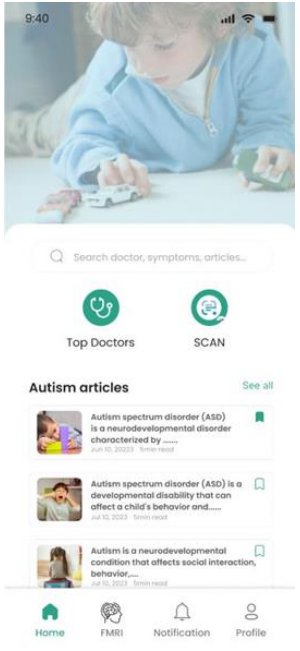
- Offline Functionality
  - Ease of access

UI/UX

Mobile User Interface

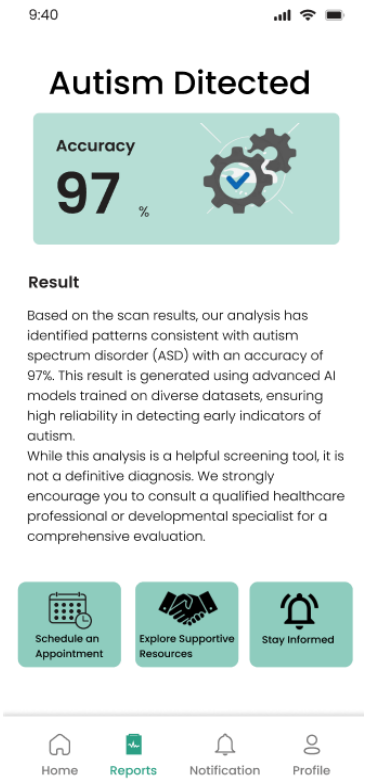






UI/UX

AUTISM DETECTED ✓



Articles

Reliable sources

- Peer-reviewed research papers.
- provide evidence-based information and resources.

Programms

Helping Activities

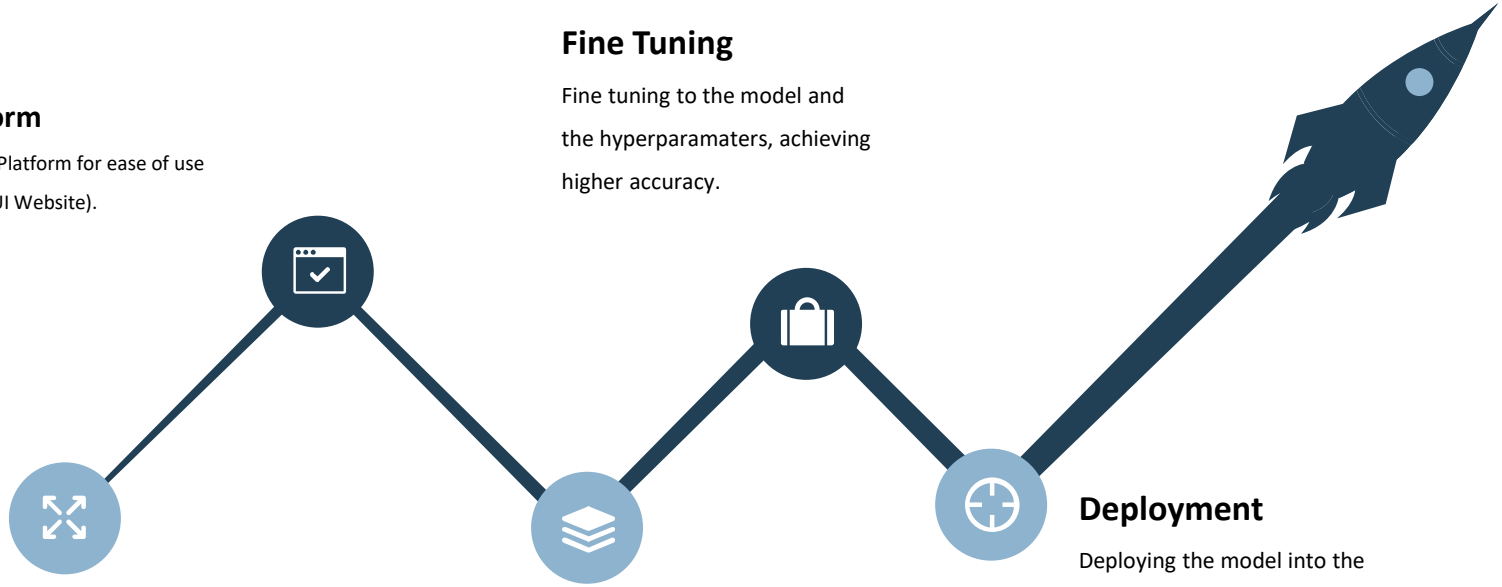
- Educational resources and materials.
- Community events.
- Support groups.

06

Future work

what we will do next.

Future Work



Platform

Provide Platform for ease of use (App, GUI Website).

Fine Tuning

Fine tuning to the model and the hyperparamaters, achieving higher accuracy.

Real-Life Data

Gather up more real-life data to test on dataset, as well as visiting the HPC to review higher resolution data and better computing power for training.

Explore Real-Life Examples

Visiting Hospitals to offer the project on an existing conditions to test and promote.

Deployment

Deploying the model into the User-Friendly Platform to be used by both Doctors and families.

THANK YOU

DO YOU HAVE ANY QUESTIONS?

Presented By:
Abdelrahman Waleed
Al-Moatasem Bellah Mohammed Mofadel
Emad Reda El-Sayed
Menna Shafik Mohamed
Momen Ahmed Mohamed Talaat El-Hofy
Shahd Hisham Abdelmonem
Shahd Usama Mastour
Tarek Mahmoud Zaki
Yusuf Waleed